Financial Econometrics

Event Study

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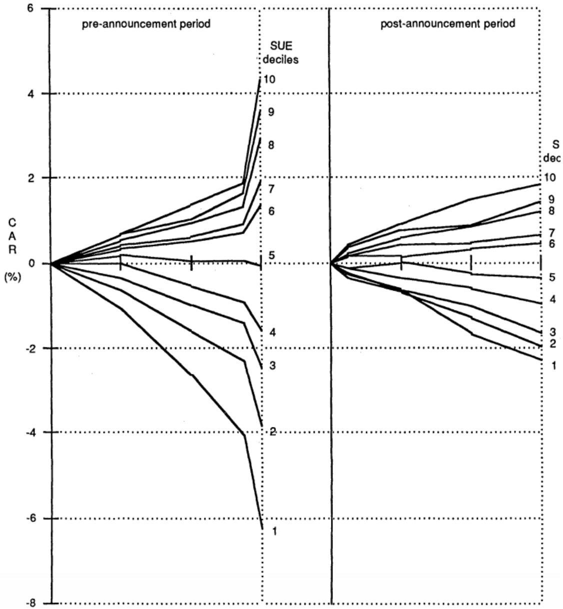
Overview

- Motivating examples
- Basics of event study
- Identification of event date
- Identification of event window
- Computing abnormal returns
- Hypothesis testing
- Alternative OLS implementation

Motivating examples

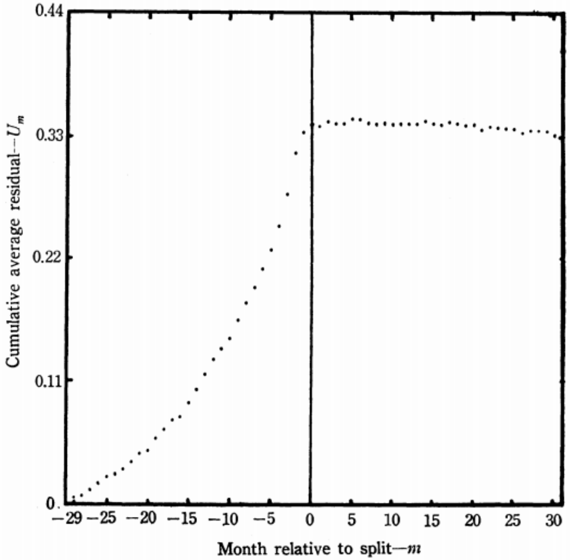
Bernard and Thomas (1989)



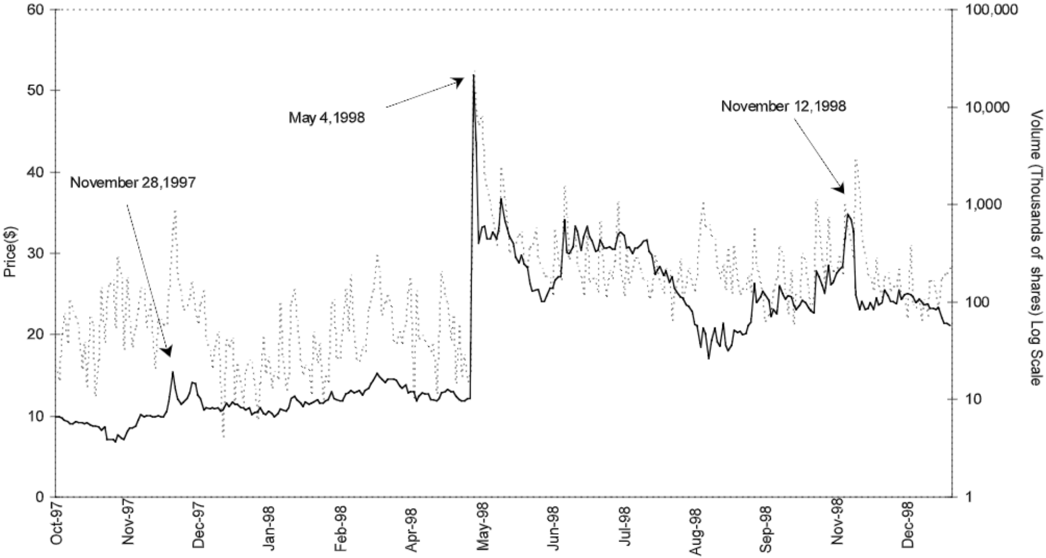
Fama, Fisher, Jensen, and Roll (1969)

FIGURE 2a

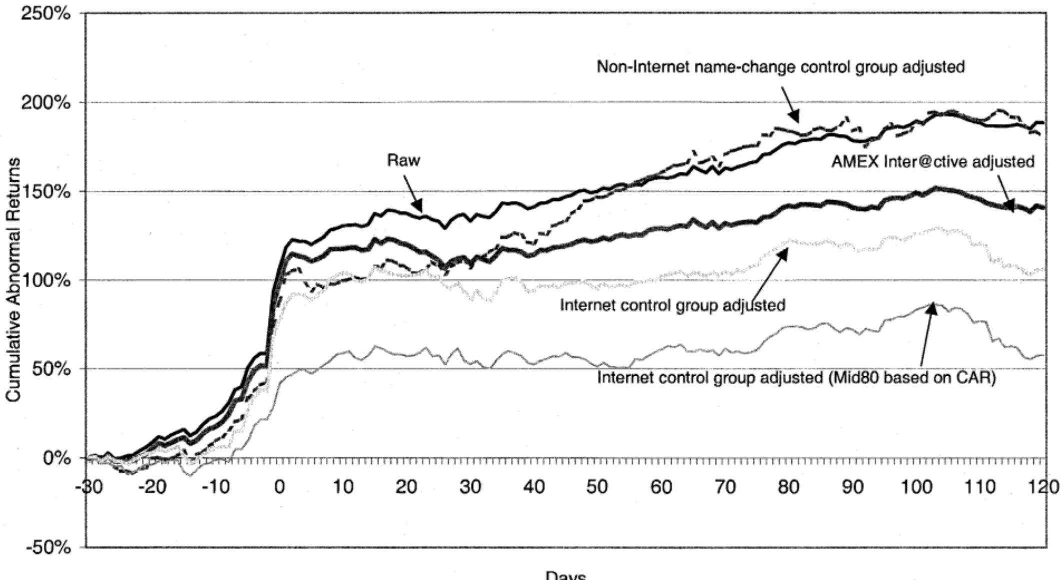
AVERAGE RESIDUALS—ALL SPLITS



Huberman and Regev (2001)



Cooper, Dimitrov, and Raghavendra (2001)



Basics of event study

Basics of event study

- Analysis of whether there was a statistically significant reaction in financial markets to past occurrences of a given type of event that is hypothesized to affect firm's market value
- Many famous results in finance derive form event studies...

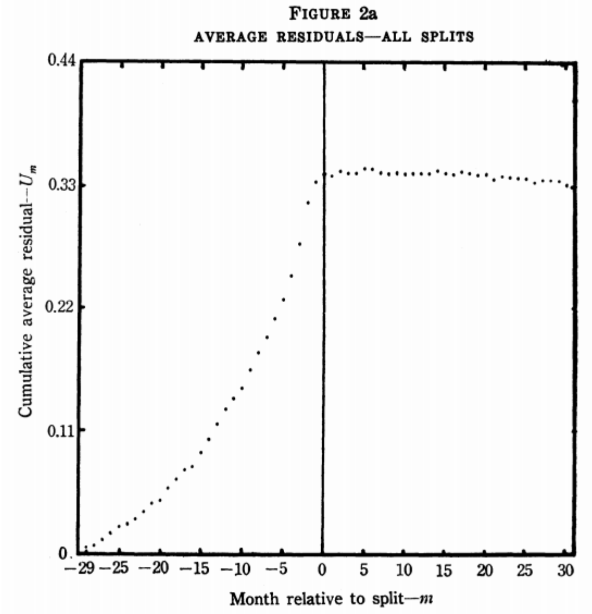
Valuable tool with two purposes

1. Price-based event studies originally designed to test the efficient market hypothesis (EMH)
   - Efficient market: market in which security price adjust rapidly to the arrival of new information and, therefore, the current prices of securities reflect all information about the security
   - Information efficiency: the speed and accuracy with which prices reflect new information
2. Value event studies emerged in 1970s
   - Aim is to examine the impact of events on the market value of specific companies, considering the market is efficient
   - Considered events: firm-specific, sector-specific, market wide, etc.

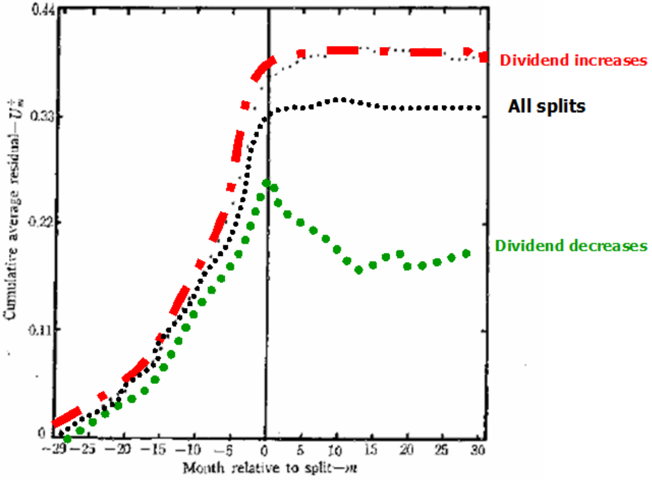
Example: Stock Splits as test of EMH

- Seminal contribution in the field is Fama, Fisher, Jensen and Roll (1969)
- They examine how stock prices adjust to the information (if any) that is implicit in stock splits
- As a natural consequence of a stock split, price per share decreases, which allows small investors to enter the market and buy stocks at a lower transaction costs
- Why stock split should convey valuable information?
  - Stock splits are usually preceded by a period of unusually high earnings (high ex post returns and stock prices)
  - Companies like to keep the dividend per share stable through time Stock split could signal that unusually high earnings will persist in the future

Example: Stock Splits as test of EMH



Example: Stock Splits as test of EMH



Implicit hypothesis of value event studies

- Financial markets are efficient and anticipation markets
  - Investors form rational expectations about future cash flows
  - In an efficient market prices are on average equal to the value of the exchanged asset
  - Recall: The value of a firm corresponds to the sum of discounted cash flows:

$$V_0 = \sum_{t=1}^{\infty} \frac{CF_t}{(1+R)^t}$$

  - The studied event impacts either future cash flows (CF) or the discount rate (R)

Quick Recap

- Motivating examples
- Basics of event study
- Identification of event date
- Identification of event window
- Computing abnormal returns
- Hypothesis testing
- Alternative OLS implementation

Event study - general procedure

- Goal: Estimate the effect of an economic event on firm value
- Crucial elements:
  1. Identification of event (nature, announcement dates, etc.)
  2. Identification of event windows
  3. Compute abnormal returns
     - Selection and estimation of normal return generating process (RGP)
  4. Compute cumulative abnormal return
  5. Hypothesis testing
     - Measure of variance

Identification of event date

Identification of event date

- You need to know the institutional details
- In particular, when is the news revealed to the market?

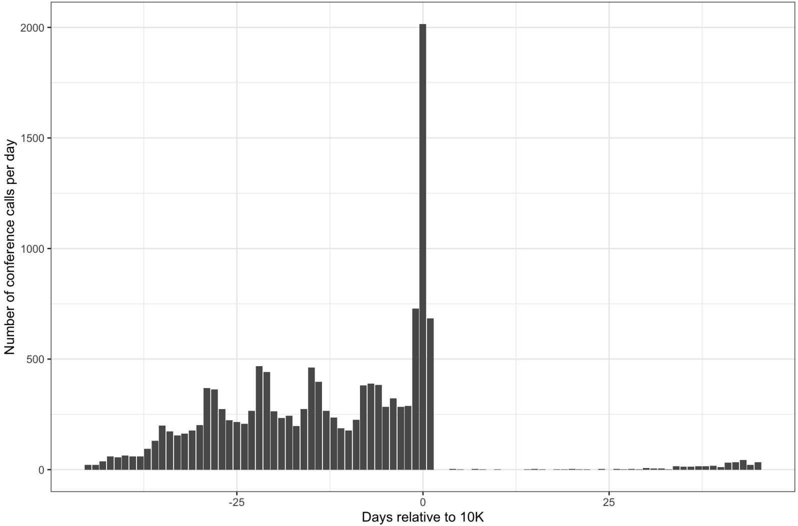
Example: Identification of event date

- For example, we are interested in how fast the information in an annual report (10-K) such as income, etc. is incorporated into the market
- Given that a 10-K includes Income Statement, Balance Sheet, and Cash Flow Statement, there seems to be a lot of relevant information
- So, we could look at return patterns around the date the annual report is published?

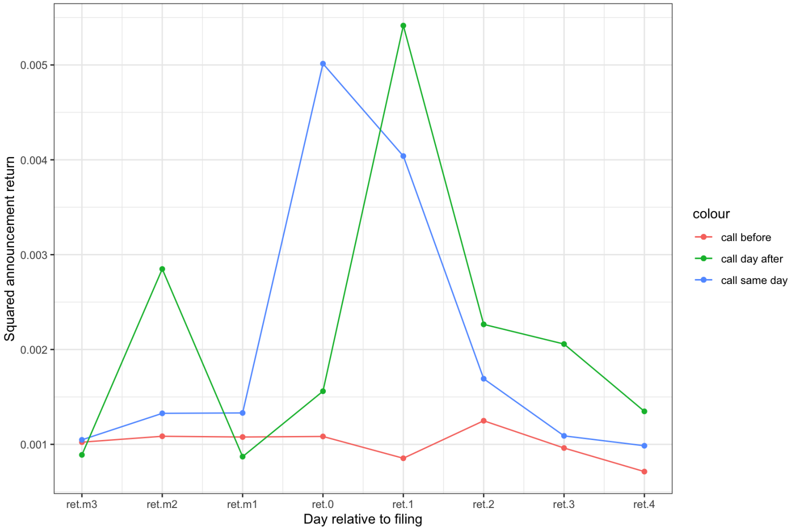
Example: Identification of event date

- But, did the market already know about the information?
- Management discusses financial results on call with analysts
- Are the calls before the annual report release date?
- Is the information revealed in the call or the annual report?

Example: Identification of event date



Example: Identification of event date



Identification of event window

Identification of event window

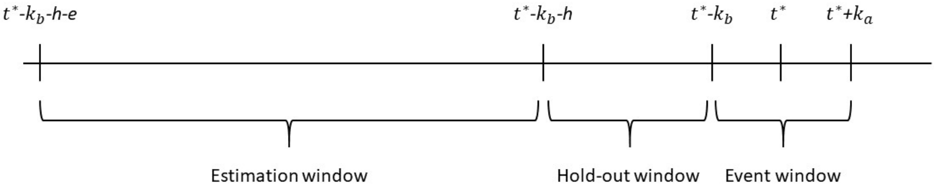


Figure 1: timeline

- Event window: Include only the event
- Hold-out: Exclude co-founding events, potential drift (information leakage)
- Estimation window: Long enough to get precise estimates, but short enough to not have structural breaks

Computing abnormal returns

Computing abnormal returns

- We are interested in the announcement return of the merger of firm A and B
- On the day of the merger, firm A' stock price fell by 2%
- Hence, the market views to merger negatively?

Computing abnormal returns

- We are interested in the announcement return of the merger of firm A and B
- On the day of the merger, firm A' stock price fell by 2%
- Hence, does the market view the merger negatively?
- On the same day, the whole market fell by 20%, and the industry of firm A experienced a total loss of 22%
- Hence, firm A performed 19-20% better than the benchmark.
- Does the market views the merger positively?
- What is the correct benchmark?

Computing abnormal returns

$$AR_{i,t} = R_{i,t} - E(R_{i,t}|X_t)$$

Computation of normal return $E(R_{i,t}|X_t)$ using return generating process (RGP) of your choice:

- Constant mean return model: $R_{i,t} = \mu_{i,t} + u_{i,t}$
- Market model: $R_{i,t} = \alpha_i + \beta_i * R_{m,t} + u_{i,t}$
- Capital asset pricing model: $R_{i,t} - R_{f,t} = \alpha_i + \beta_i * (R_{m,t} - R_{f,t}) + u_{i,t}$
- Fama-French model: $R_{i,t} - R_{f,t} = \alpha_i + \beta_{mkt} * (R_{m,t} - R_{f,t}) + \beta_{SMB} * SMB_t + \beta_{HML} * HML_t + u_{i,t}$
- etc.

Computing cumulative abnormal return

Single event: cummulative abnormal return (CAR) defined as abnormal return cummulated between day K (1st day of the event window) and day L (last day of the event window)

$$CAR(K,L)_i = \sum_{t=K}^{L} AR_{i,t}$$

Multiple event: cummulative average abnormal return (CAAR) defined as average of firm individual CARs

$$CAAR(K,L) = \sum_{i=1}^{N} CAR(K,L)_i$$

Buy-and-hold return

- Another option is buy-and-hold returns (BHR)

$$BH(K,L)_i = \prod_{t=K}^{L} (1 + AR_{i,t}) - 1$$

- If r is small, CAR and BH returns are approximately equal.

Hypothesis testing

Hypothesis testing

Single event:

- H0: $CAR_i = 0$ vs. $CAR_i \neq 0$
- $t = \frac{CAR(K,L)_i}{\sqrt{k * \hat{\sigma}_i^2}}$ with k = length of the event window
- $\hat{\sigma}_i^2$ computed on estimation window. Use variance of error term of RGP $u_{i,t}$

Mutliple event:

- H0: $CAAR = 0$ vs. $CAAR \neq 0$
- $t = \frac{CAAR(K,L)}{\sqrt{k * \hat{\bar{\sigma}}^2}}$
- with $\hat{\bar{\sigma}}^2 = \frac{1}{N^2} \sum_{i=1}^{N} \hat{\sigma}_i^2$

Caveat

- Variance during events higher than in normal times, hence t-value might be overstated
- Cross-sectional standard standard error

$$CAR_i = \alpha + u_i$$

Use of CAR and BHR in other circumstances

Often we are interested whether there is an association between abnormal return and characteristica of a single event:

- Cross-sectional: e.g. acquirer announcement return and the type of surprise, method of payment, etc.
- Panel-data: earnings announcement return and the type of surprise, etc.

Sometimes we want to control for past returns. E.g. Runnup return to a merger.

Alternative OLS implementation

Alternative OLS implementation

- Use estimation, hold-out, and event window and estimate RGP including a dummy variable indicating event day.
- For example, using CAPM:

$$R_{i,t} - R_{f,t} = \alpha_i + \gamma I(t = Event\ day)_t + \beta_i * (R_{m,t} - R_{f,t}) + u_{i,t}$$

- If you have a multi-day event window you can include several dummies for each day (as in the assignment), or one dummy turning one if during the whole event window
- Run individually for each firm, and use $\hat{\gamma}$ in cross-section/panel

Summary

- Analysis of return reaction to an event
  - Event date
  - Event window
  - Abnormal return
  - Variance
- Use in alternative circumstances
- Alternative OLS implementation